Exhibits (tables and figures) for: Revealing hidden jump strategies beyond conventional force-plate metrics

Tables

Table 1. Sample characteristics

				Quartiles		Range	
Variable	Mean	SD	Median	Q1	Q3	Min	Max
Athletes (n = 212)							
Age at first test (years)	16.0	5.9	14.1	12.0	16.8	8.8	37.9
Body mass at first test (kg)	53.3	15.9	51.7	40.9	64.0	25.0	96.0
Trials per athlete	3.8	2.3	3.0	2.0	5.0	1.0	16.0
Sessions per athlete	1.5	0.8	1.0	1.0	2.0	1.0	6.0
Sessions (n = 327)							
Trials per session	2.5	0.8	2.0	2.0	3.0	1.0	7.0
Trials (n = 805)							
Jump height (cm)	28.2	8.4	27.0	21.6	34.7	8.5	51.5
Concentric peak force (BW)	2.35	0.38	2.29	2.11	2.51	1.65	4.70
Eccentric braking impulse (N s)	27.8	15.4	24.6	16.6	35.7	0.0	95.2
Concentric impulse, first 100 ms (N s)	62.2	26.7	58.7	41.8	79.1	1.5	187.9
Peak landing force (BW)	5.39	1.97	4.97	3.88	6.67	2.19	14.88
Take-off duration (s)	0.78	0.19	0.77	0.66	0.90	0.31	1.48
Landing duration (s)[a]	0.25	0.11	0.24	0.15	0.31	0.07	0.93
Native samples, take-off[b]	755	212	762	627	889	180	1484
Native samples, landing[b]	238	116	223	144	310	41	928

Note. BW = multiples of body weight. Take-off runs from movement onset to take-off (vendor contraction time). 59 trials were recorded at 500 Hz and 746 at 1000 Hz. [a]Touch-down to the lowest point of the centre of mass. [b]Phase duration multiplied by the sampling rate; each phase is resampled to 101 nodes.

Table 2. Where along each phase the predictors are associated with force (function-on-scalar regression)

Predictor	Simultaneous 95% band excludes zero (% of phase)			
	Athlete bootstrap	Mixed model	Both (claimed)	Largest effect, BW per SD [95% CI][a]
Take-off (onset to take-off)				
Jump height	13–33, 65–99	4–30, 38–68, 72–84, 88–98	14–30, 66–68, 72–84, 88–98	0.25 [0.22, 0.29] at 95%
Age	0, 95–99	76–90, 98	98	0.09 [0.06, 0.13] at 97%
Body mass	0–6, 98–100	2–14, 42–66, 94–100	2–6, 98–100	−0.07 [−0.15, 0.01] at 78%
Jump height, within athlete[b]	3–29, 46–63, 92–99	–	–	0.029 [0.017, 0.042] per cm at 58%
Jump height, between athlete[b]	18–29, 67–99	–	–	0.031 [0.026, 0.035] per cm at 95%
Landing (touch-down to lowest centre of mass)				
Jump height	3–28	0–32, 80–100	4–28	0.36 [0.21, 0.52] at 14%
Age	none	none	none	0.34 [0.06, 0.62] at 35%
Body mass	0, 75–83	0–100	0, 76–82	−0.37 [−0.63, −0.11] at 33%
Jump height, within athlete[b]	2–16, 23	–	–	0.115 [0.074, 0.164] per cm at 13%
Jump height, between athlete[b]	9–15, 18–25	–	–	0.039 [0.016, 0.061] per cm at 24%

Note. Model: force ~ jump height + age + body mass, predictors standardised (1 SD = 8.4 cm, 5.7 years, 15.8 kg). Athlete bootstrap: pointwise least squares with 2000 resamples of athletes. Mixed model: penalised functional regression with athlete and session random curves, fitted on every second node. [a]Least-squares estimate with pointwise 95% bootstrap interval. [b]Jump height split into the athlete's mean and the trial's deviation from it; effects per cm with 95% bootstrap intervals.

Table 3. Pointwise group comparisons of athlete-mean curves with family-wise error control

Contrast (athletes)	Phase	Significant region, % (p)	Difference at peak, BW [95% CI]	Hedges' g [95% CI]	Region after adjustment, % (p)[a]
Top vs bottom jump-height tertile (71 vs 71)	Take-off	81–99 (<0.001)	0.55 [0.48, 0.61] at 93%	2.63 [2.31, 2.94]	12–32 (0.021); 66–99 (<0.001)
	Landing	2–56 (0.003)	0.75 [0.45, 1.04] at 15%	0.83 [0.50, 1.15]	19–40 (0.043)
18 and over vs under 18 (48 vs 164)	Take-off	87–99 (<0.001)	0.47 [0.36, 0.56] at 96%	1.60 [1.25, 1.92]	none (min p = 0.12)
	Landing	none (min p = 0.07)	–	–	none (min p = 0.23)

Note. One mean curve per athlete. Clusters of adjacent nodes with $|t|$ above the two-sided 5% threshold were tested against the permutation distribution of the largest cluster mass over the whole phase (5000 permutations of athlete labels), which controls the family-wise error rate over the 101 nodes. Differences are first group minus second, at the node of largest $|t|$, with 95% intervals from 2000 bootstrap resamples of athletes within groups. Bonferroni threshold across the four primary tests: $p < 0.0125$. [a]Tertile contrast adjusted for athlete-mean age and body mass; age contrast adjusted for body mass and jump height (Freedman–Lane permutation).

Table 4. Curve profiles: description, validation and stability within athletes

	Take-off	Landing
Description		
Profile 1 / profile 2 (trials)	329 / 476	560 / 245
Mean jump height (cm)	29.6 / 27.2 ($p < 0.001$)[a]	28.4 / 27.7 ($p = 0.017$)[a]
Mean take-off duration (s)	0.65 / 0.87	0.79 / 0.76
Mean landing duration (s)	0.24 / 0.25	0.29 / 0.14
Is there cluster structure?		
Mean silhouette	0.30	0.37
Number of groups preferred by the gap statistic[b]	1	6
Dip test of unimodality, first FPC score (p)	0.74	0.98
Bootstrap Jaccard similarity (range over profiles)	0.93–0.95	0.97–0.99
Stability		
Same profile in every session, % (athletes with ≥ 2 sessions, $n = 83$)	55	70
Expected by chance, % (permutation p)	43 (0.007)	51 (<0.001)
Two trials in one session share a profile, % (chance)	76 (52)	80 (58)
Profile mix in the matched-height band, % (all trials)[c]	35 / 65 (41 / 59)	74 / 26 (70 / 30)
Agreement with profiles after elastic alignment (ARI)	–	0.17

Note. Profiles from k-means ($k = 2$) on functional principal component (FPC) scores covering at least 90% of variance, computed separately for each phase. [a]Linear mixed model with an athlete random intercept. [b]Candidates 1 to 6; 1 means no cluster structure. [c]Trials with jump height between the 45th and 55th percentiles (26.0–28.2 cm, 82 trials).

Figures

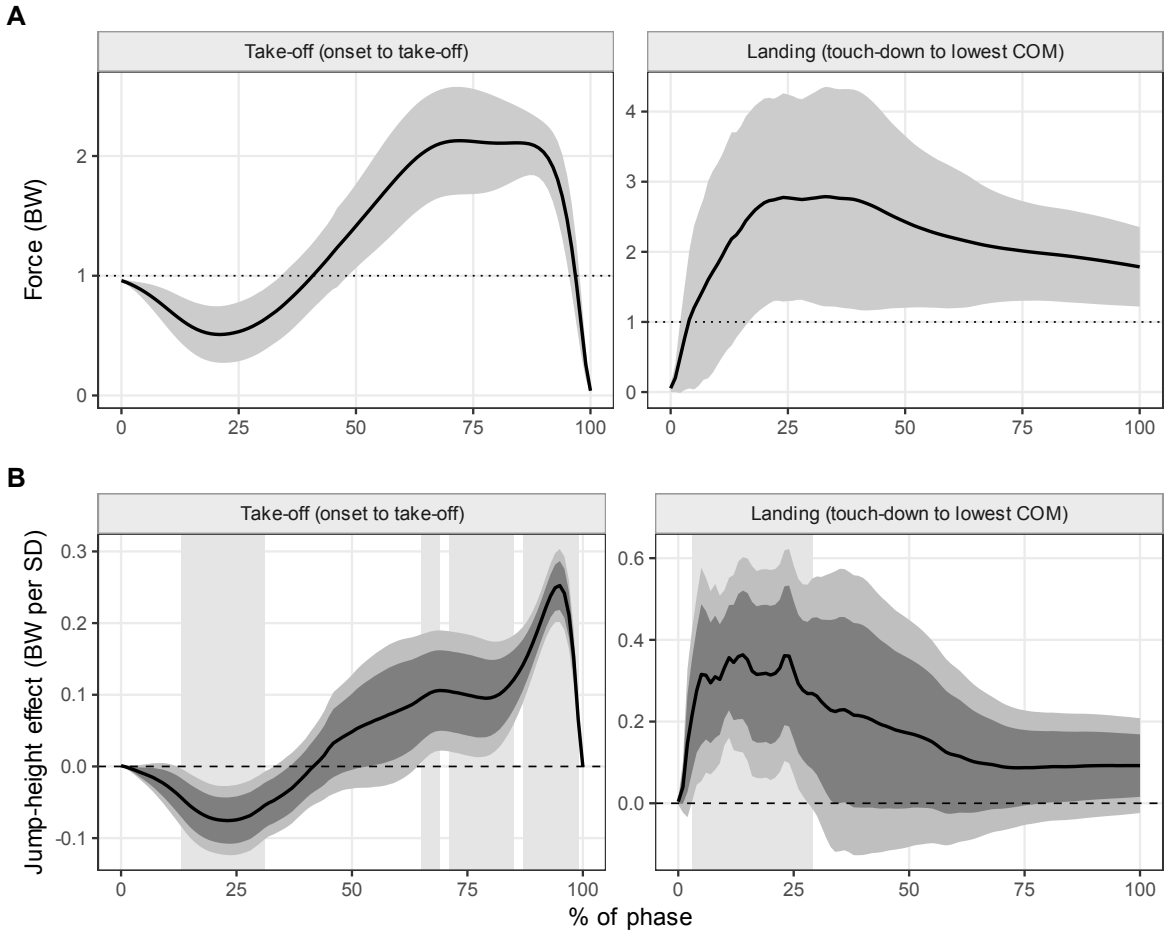


Fig. 1. Average curves and where jump height matters. (A) Mean force in multiples of body weight (BW) over the 805 analysed trials (line) with ± 1 SD (shaded); the dotted line is body weight. Take-off runs from movement onset to take-off and landing from touch-down to the lowest centre-of-mass (COM) position, each rescaled to 0–100%. (B) Jump-height coefficient from function-on-scalar regression (BW per SD of jump height, adjusted for age and body mass) with pointwise (dark) and simultaneous (light) 95% bands from resampling athletes. Pale background shading marks the regions where both estimation approaches agree.

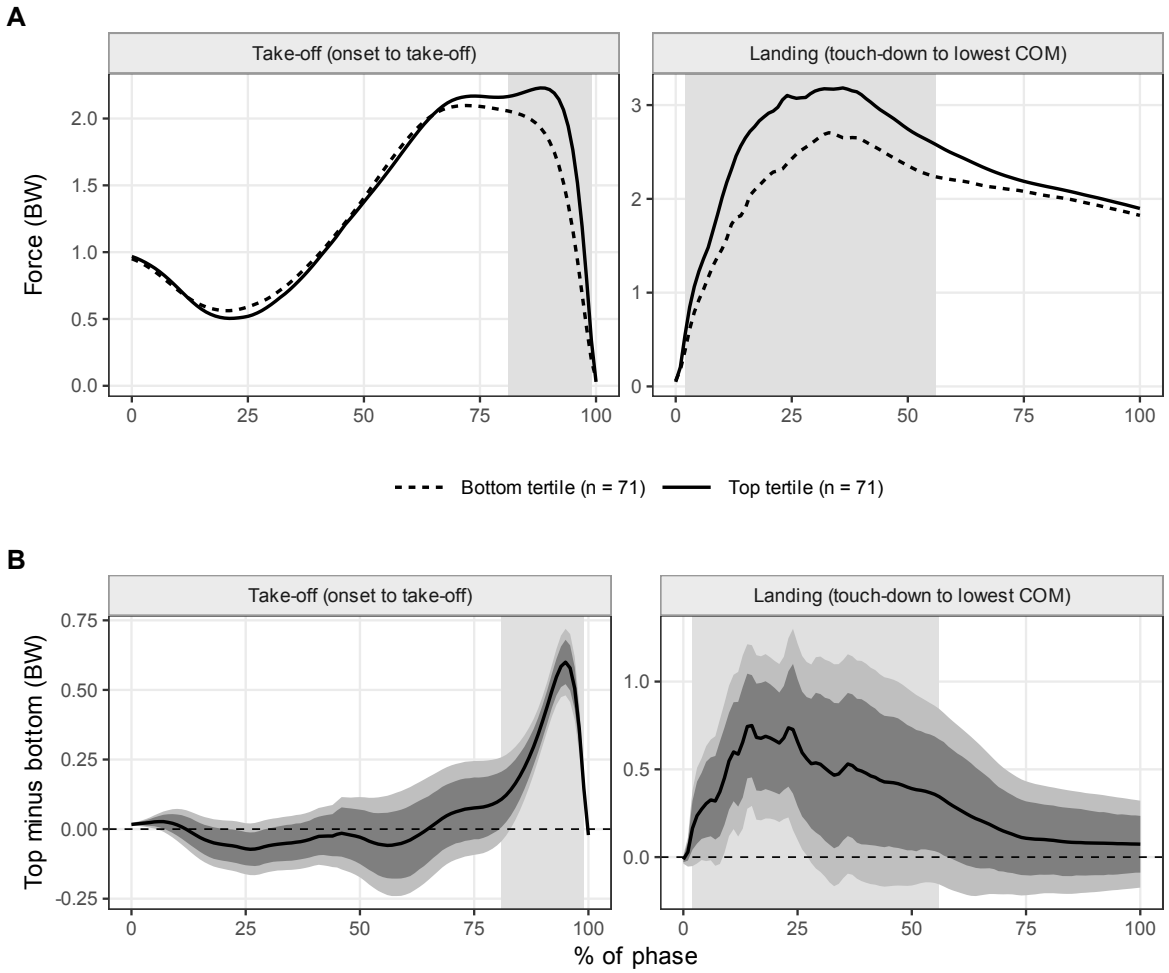


Fig. 2. Top versus bottom jump-height tertile, one mean curve per athlete. (A) Group mean curves. (B) Difference (top minus bottom) with pointwise (dark) and simultaneous (light) 95% bootstrap bands. Background shading marks clusters significant in the permutation test with family-wise error control.

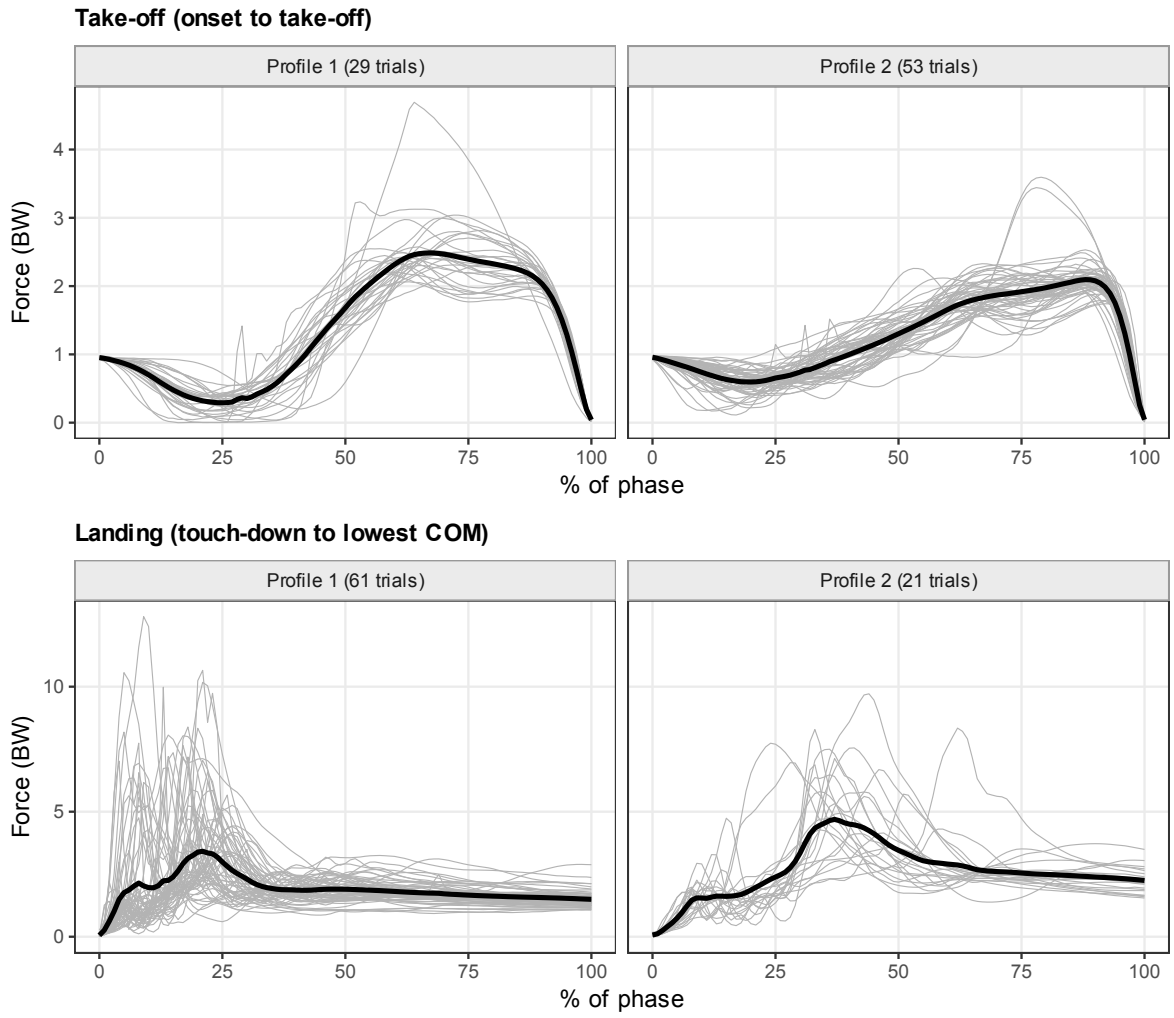


Fig. 3. Jumps of almost equal height (26.0–28.2 cm, 82 trials) still differ in shape. Thin grey lines are individual trials, split by curve profile; the thick black line is the profile mean. Top: take-off. Bottom: landing.

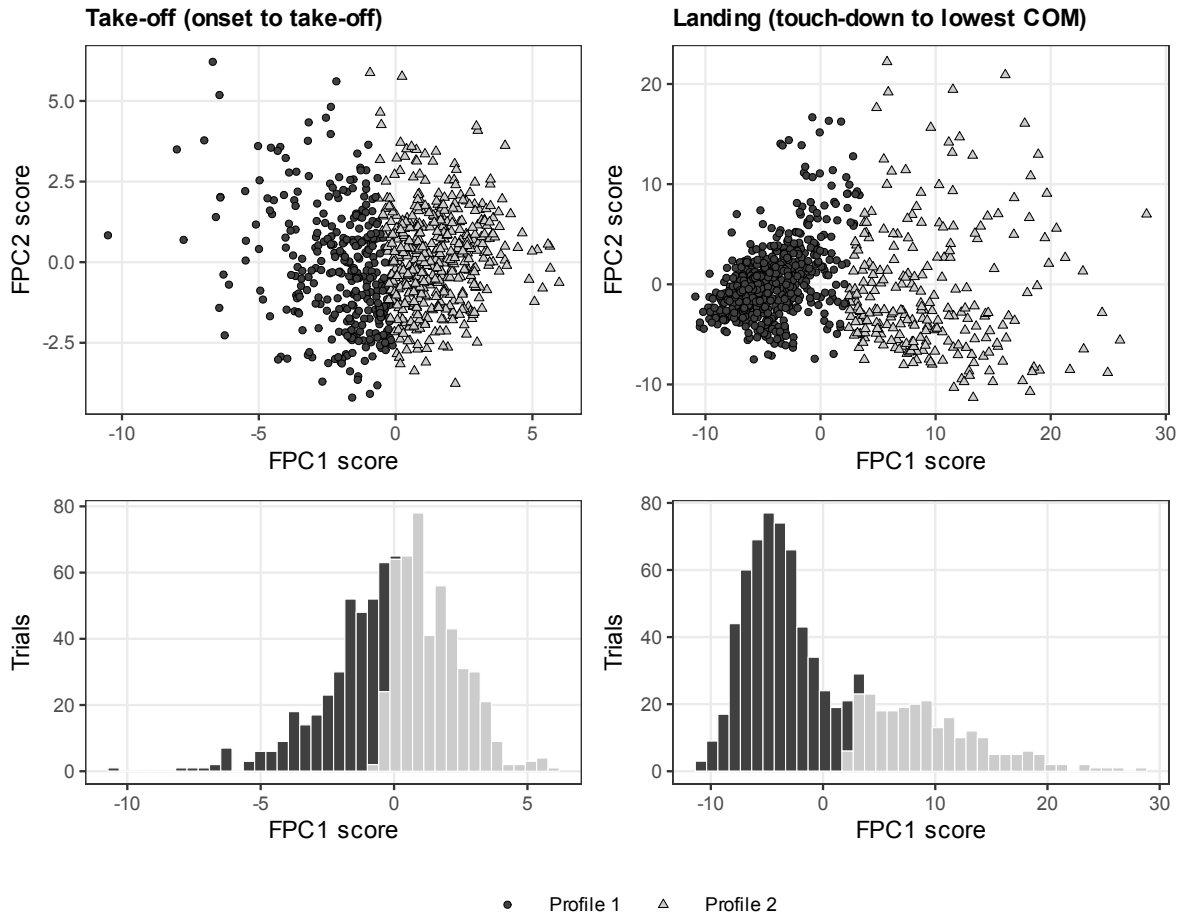


Fig. 4. Curve profiles form a continuum. For each phase, the scatter plot shows the first two functional principal component (FPC) scores of every trial, with symbols for the two k-means profiles, and the histogram shows the distribution of the first score shaded by profile. There is no gap between the profiles and no second mode.